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A federated learning system with enhanced feature extraction for human activity recognition

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ABSTRACT

With the rapid growth of mobile devices, wearable sensor-based human activity recognition (HAR) has become one of the hottest topics in the Internet of Things. However, it is challenging for traditional approaches to achieving high recognition accuracy while protecting users' privacy and sensitive information. To this end, we design a federated learning system for HAR (HARFLS). Based on the FederatedAveraging method, HARFLS enables each user to handle its activity recognition task safely and collectively. However, the recognition accuracy largely depends on the system's feature extraction ability. To capture sufficient features from HAR data, we design a perceptive extraction network (PEN) as the feature extractor for each user. PEN is mainly composed of a feature network and a relation network. The feature network, based on a convolutional block, is responsible for discovering local features from the HAR data while the relation network, a combination of long short-term memory (LSTM) and attention mechanism, focuses on mining global relationships hidden in the data. Four widely used datasets, i.e., WISDM, UCI_HAR 2012, OPPORTUNITY, and PAMAP2, are used for performance evaluation. Experimental results demonstrate that PEN outperforms 14 existing HAR algorithms on these datasets in terms of the F1-score; HARFLS with PEN obtains better recognition results on the WISDM and PAMAP2 datasets, compared with 11 existing federated learning systems with various feature extraction structures.

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1. Introduction

Over the years, the Internet of Things (IoT) has shown huge potential to promote social development on a variety of aspects [1,2], for instance, mobile networks [3], blockchain [4], autonomous driving [5], and software defined networking [6]. Among them, human activity recognition (HAR) is nowadays one of the most active topics [7]. HAR aims to identify the actions carried out by a person, based on a set of observations on himself/herself and the corresponding surrounding environment [8]. HAR has been applied to various application areas, including surveillance system [9], behavior analysis [10], electroencephalography [11], gesture recognition [12], and healthcare system [13].

In the past two decades, a large number of algorithms were proposed to address various HAR problems [14–16]. These algorithms are roughly classified into two categories, namely vision-based and sensor-based approaches [17]. The former is primarily based on computer vision techniques that recognize camera captured data. Although they may achieve decent performance, the vision-based approaches face a number of non-trivial challenges. For example, the process of image data collection may expose one's privacy. These data are also easily affected by environmental factors, including light, background, and pixels. On the other hand, increasingly more research efforts have been dedicated to sensor-based approaches, thanks to the rapid development in sensing technology [18]. A sensor-based approach usually extracts a timely ordered sequence from a set of linear or angular velocity points collected by sensors [19,20] and captures features directly from this sequence by a feature extraction algorithm.

The sensor-based approaches can be divided into three categories, namely wearable, object-tagged (device-bound), and environment-tagged (device-free/ dense sensing) [17,21]. Because of the popularity of mobile devices, e.g., smartphones and smartwatches, collecting data by wearable sensors becomes increasingly more realistic and convenient. A number of learning structures have been developed for wearable sensor-based activity recognition, e.g., multi-head convolutional attention [22], autoencoder [23] and layer-wise convolutional neural networks (CNN) [24]. For the object-tagged approaches, sensors are attached to objects of a certain scene. When these objects are used, the corresponding sensors are activated to collect scenic data. In autonomous driving, for example, lane sensors in the car are responsible for collecting real-time road condition information on

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lane monitoring. In addition, an environment-tagged approach does not require users to carry any tag or device for capturing their motions and gestures [17,21]. Radio frequency identification (RFID) is one of the widely recognized environment-tagged techniques for HAR, e.g., human body fall detection [25] and occupancy detection [26].

No matter which category above it belongs to, a sensor-based HAR algorithm aims to gain high recognition accuracy. In general, such an algorithm is deployed either on a local client or on a central server, where user data are analyzed and processed. The HAR performance is significantly affected by the motion data coming from one user for those deployed on individual clients. Therefore, it is natural to use the motion data collected from as many users as possible to ensure high HAR accuracy. An external service makes this possible by encouraging all users to upload their data to a server on which these data are used to ensure the integrity of data analysis. However, the uploaded data may contain privacy and sensitive information of users, e.g., GPS locations and bank accounts, causing personal safety and property security problems if data leakage happens. Thus, increasingly more research attention has been paid to protecting the procedures of private data analysis [27]. Some researchers aim at removing as much sensitive information as possible from the uploaded data, such as the privacy-preserving data-leak detection [28] and the obfuscation-resilient privacy leak detection [29]. However, it is unlikely to remove all sensitive data from the uploaded data. Besides, sensitive information removal is not beneficial to avoid exposing one's personal activity information to others. To this end, supervising the data uploading and analysis processes is also necessary, e.g., multi-party classification protocol [30]. But, these supervision processes are computationally expensive since they incur non-trivial hardware resource consumption on servers.

To overcome the difficulties above, McMahan et al. [31] introduced a novel distributed learning technique, referred to as federated learning (FL), which allows users to collectively harvest the benefits of shared models trained from their local data without sending their data to others. In this way, users' privacy and sensitive information is protected while promising data analysis is achieved. FL has been viewed as an emerging architecture to support a variety of privacy-sensitive domains [32,33], such as cloud computing [34], mobile edge computing (MEC) [35], health analysis [36], mobile crowdsourcing [37], and emotion prediction [38].

Unlike the object- and environment-tagged methods, it is easier for the wearable approaches to collect sufficient data of human activities, which is more conducive to identifying human activities and improving the quality of our daily life, thanks to the widespread popularity of mobile devices [39]. This paper focuses on wearable sensor-based HAR problems and presents a federated learning system (HARFLS). HARFLS consists of a central server and a number of mobile users running identical HAR application. The server collects the model weights from all users and distributes the average weights to each user for the model update, where any user never uploads its raw data generated locally to the server. This significantly decreases the communication cost incurred in the distributed system and effectively reduces the data leakage risks.

Specifically, our main contributions are listed as follows:

- We design a federated learning system for wearable sensor-based HAR, namely HARFLS. It uses the FederatedAveraging method for model weights sharing and a perceptive extraction network (PEN) for feature extraction. In addition, the widely recognized homomorphic encryption technique is applied to reduce the data leakage risks during weights uploading and distribution.
- PEN is deployed on each mobile device as its feature extractor. It consists of a feature network and a relation network. These two networks primarily rely on one-dimensional CNN for local feature extraction and long short-term memory-based attention for global relationship discovery. So, PEN can extract rich and hierarchical representations from the HAR data.
- Experimental results show that PEN performs better than 14 HAR algorithms on the wireless sensor data mining (WISDM), University of California Irvine (UCI) HAR using smartphones in 2012 (UCI_HAR 2012), OPPORTUNITY, and PAMAP2 datasets, and outperforms 11 existing FL feature extractors in terms of the F1-score on the WISDM and PAMAP2 datasets with millions of samples. We study the impacts of the numbers of users and connected users on the HARFLS's performance, reflecting the relationship between recognition accuracy and system capacity. Also, HARFLS overweighs 11 FL systems regarding the F1-score on the WISDM and PAMAP2 datasets.

The rest of the paper is organized as follows: Section 2 reviews the related work. Section 3 introduces the proposed feature extractor, i.e., PEN, and the federated system, i.e., HARFLS. The experimental results are analyzed and discussed in Section 4. Finally, conclusions are drawn in Section 5.

2. Related work

This section reviews the wearable sensor-based HAR approaches and federated learning techniques, respectively.

2.1. Wearable approaches for HAR

So far, wearable sensor-based HAR problems have attracted significant research attention from both academia and industry [15,16,40]. To address them, a large number of HAR algorithms have been proposed [17]. There are roughly two research streams, including traditional algorithms and deep learning ones [5,7]. The former is only able to extract shallow features from data [41], including, k-nearest neighbor (KNN), support vector machine (SVM), Bagging, logistic regression (LR), Naïve Bayes, centroid-distance-vector collaborative learning, etc [42–45]. These algorithms usually lead to low HAR accuracy.

Unlike the traditional algorithms, the deep learning (DL) ones aim to unveil the representation hierarchy of data [46], which helps to obtain both basic and intricate features and thus higher accuracy. Alsheikh et al. [47] hybridized DL and hidden Markov models for efficient representation extraction. In [42], a temporal convolutional model on low-power devices achieved decent performance on the WISDM dataset. Gu et al. [23] designed a stacked denoising autoencoder to handle the data collected from multiple sensors. Keshavarzian et al. [48] proposed a modified deep residual network for HAR, which achieved decent performance on the UCI datasets. In [22], a multi-head convolutional attention approach was used to recognize six human activities, including walking, jogging, upstairs, downstairs, sitting, and standing. Tang et al. [24] put forward a lightweight CNN with Lego filters to strike a balance between accuracy and efficiency. Wijekoon et al. [49] used a personalized open-ended algorithm based on a matching network to explore the latent information of input. In [50,51], a cascading model of the CNN and LSTM networks was developed to mine temporal features from HAR data.

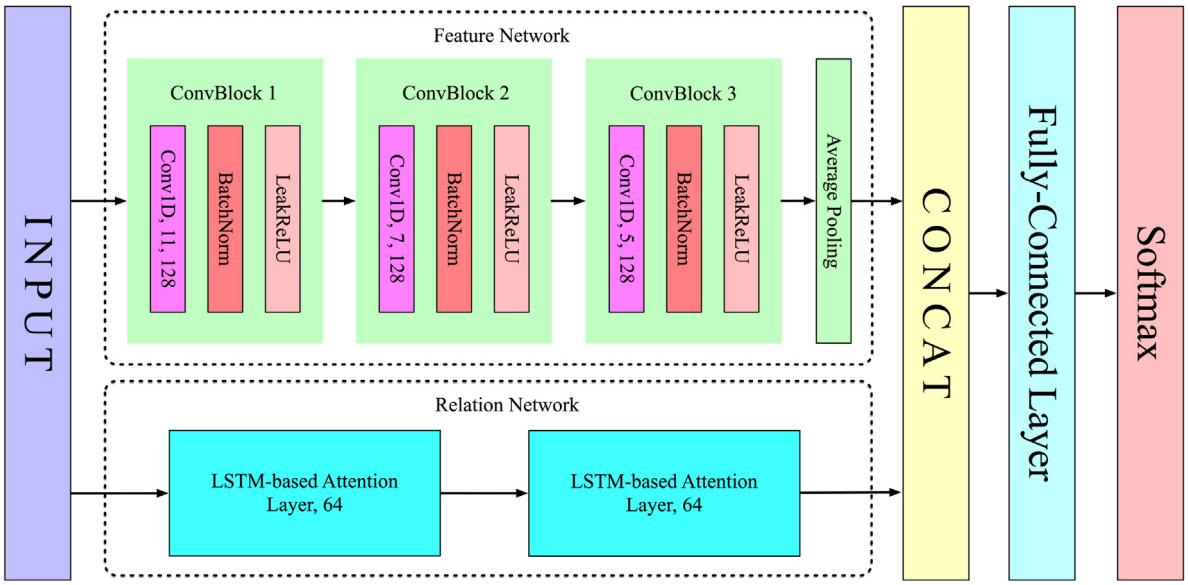


Fig. 1. The overview of PEN. Note that 'Conv1D, 11, 128' represents a 1-dimension convolutional neural network, where its filter size and channel sizes are set to 11 and 128, respectively; 'BatchNorm' is a batch normalization module; 'LeakReLU' is the leaky rectified linear unit activation function; and 'LSTM-based Attention Layer, 64' is an LSTM-based attention layer with 64 units.

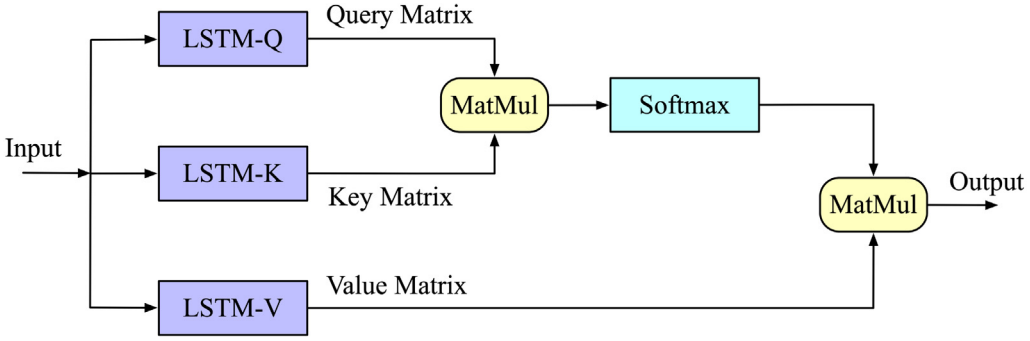


Fig. 2. Structure of the LSTM-based attention layer [52].

2.2. Federated learning

Let $\boldsymbol{U} = \{U_1, U_2, \ldots, U_{N_{tot}}\}$ be the set of users in a federated learning system, where N_{tot} is the number of users and U_i is the ith user. User U_i is associated with a local data D_i and a local model M_i, $i = 1, 2, \ldots, N_{tot}$. U_i uses D_i to train its local model M_i and shares this model with the FL system. The system merges the models coming from various users via some concatenated methods, namely $M_{sum} = Converge(M_1, M_2, \ldots, M_{N_{tot}})$, and distributes M_{sum} to each user for local model update, where *Converge* is a converging function. Note that the concatenated method plays an important role in the FL system [31]. An appropriate one helps to facilitate system performance. The FederatedAveraging (FedAvg) method is one of the commonly used concatenated methods [33]. This method is to calculate the average weights of the models of all users in the FL system. Compared with other concatenated methods, e.g., the gradient-based concatenation [32], the FedAvg method is simpler and more efficient. For example, Huang et al. [36] used the FedAvg method to converge the models collected from 50 hospitals, achieving decent performance in mortality and stay time prediction. In [53], the FedAvg method is integrated into an edge computing system to handle computation offloading. Meanwhile, this method has been widely applied to emoji prediction [38], prognostics and health management [54], medical image classification [55], HAR [56–58], and so on.

Addressing HAR problems by FL techniques has attracted more research attention increasingly. For example, Zhao et al. [56] presented an FL system that used mobile devices with long short-term memory (LSTM) networks to monitor activity and health condition locally, where the number of users was fixed to 100. In [57], the authors evaluated a number of FL aggregation algorithms for HAR and showed the one with two-layer CNN-based model achieved decent performance, where 15 users collectively updated the global model. However, in the real world, the number of users connected to the FL system is not fixed but changing dynamically. To study an FL scenario with a fixed number of users is not that practical. Some researchers hence considered scenarios with different numbers of users, e.g., Wu et al. [58] proposed a personalized FL framework based on CNN and investigated the impact of 3, 5, 10, 20, and 30 users on the recognition accuracy. The authors assumed that all users in the system are active and contribute to the global model update. This was, however, still not practical since the network connectivity and the user activity were ignored.

2.3. Analysis and motivation

Currently, most of the existing DL-based HAR algorithms, e.g., the multi-head convolutional attention [22], Lego CNN [24], and LSTM- and CNN-based feature extractors in FL-enabled HAR

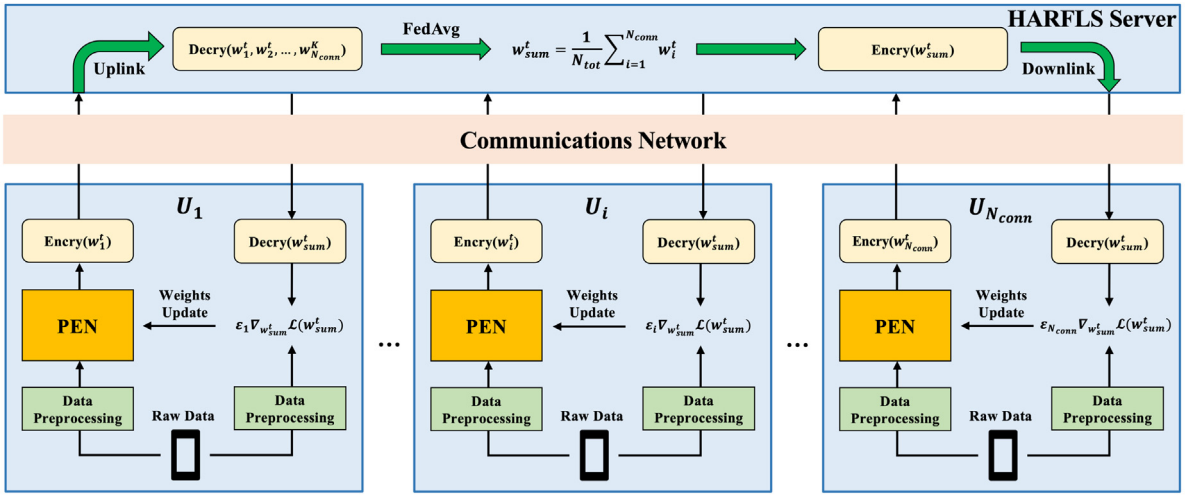


Fig. 3. The architecture of HARFLS. Note that 'Encry' and 'Decry' are the encryption and decryption functions of the homomorphic encryption, respectively.

[56–58], use one (usually hybrid) network for feature and relation extraction. However, these algorithms are more active in exploring the local patterns than capturing the global ones of HAR data, which usually results in deteriorated performance. This motivates the authors to construct a dual-network-based architecture for feature and relation extraction, i.e., the proposed perceptive extraction network (PEN). PEN consists of two parallel networks: a feature network, and a relation network. The feature network is responsible for extracting the local patterns of data while the relation network focuses on capturing the global patterns of data. In this way, sufficient features and relationships result, which is beneficial to effectively recognize human activities.

3. Methodology

This section first introduces PEN and then overviews the architecture of HARFLS.

3.1. Perceptive extraction network (PEN)

Feature extraction is of vital importance in FL systems since information of human activities is hidden in time series data. Various deep neural networks have been adopted to extract HAR patterns, e.g., LSTM [56] and CNN [57,58]. However, these deep structures cannot take care of local feature extraction and global relationship extraction simultaneously, as explained in Section 2.2 (Related Work). This is our motivation to devise PEN, a comprehensive feature extractor for HARFLS. PEN is mainly composed of a feature network and a relation network, as shown in Fig. 1. The feature and relation networks are responsible for exploring sufficient local features and global relationships from HAR data.

3.1.1. Feature network

In the well recognized FCN-LSTM structure, utilizing three fully-connected layers results in promising performance when extracting local temporal features [59]. Inspired by this idea, we design the feature network consisting of three convolutional blocks. In each convolutional block (ConvBlock), there is a 1-dimensional CNN (Conv1D) module, a batch normalization (BatchNorm) module [60] and a leaky rectified linear unit (LeakReLU) activation function.

The Conv1D module, as the core component of ConvBlock, is used to mine the local features from the input data. BatchNorm helps to facilitate the training efficiency by getting rid of the internal covariate shift [60]. Meanwhile, LeakReLU helps to reduce the loss of features during their transmission process by taking both positive and negative numbers into account.

3.1.2. Relation network

Recently, a two-layer LSTM-based attention network obtained decent accuracy results on a variety of univariate and multivariate time series datasets [52]. Therefore, we incorporate the attention network above into PEN as the relation network. The first and second layers are responsible for extracting the basic relationships from HAR data and mining the intricate connections among the relationships previously obtained, respectively (see Fig. 2).

In each LSTM-based attention layer, LSTM networks are embedded into an attention structure, as illustrated in Figure 2, where 'MatMul' denotes a matrix multiplication operation. This layer maps a temporal query and a set of key–value pairs to an output, $O_{LSTMAtt}$, where the query, *Query*, key, *Key*, and value, *Value*, are the feature matrices captured by the three LSTM networks [52,61]. $O_{LSTMAtt}$ is defined in Eq. (1).

$$O_{LSTMAtt} = Softmax(Query \cdot Key^T) \cdot Value \tag{1}$$

where, *Softmax* is a widely adopted function to compute the possibilities of a given matrix, and Key^T is the transpose of *Key*.

3.1.3. Loss function

As shown in Fig. 1, a concatenated layer is used to merge the local features and global relationships previously extracted into representations. The fully-connected layer and *Softmax* play a classifier role that categorizes these representations into C classes, where C is the number of categories. Let y_j and p_j denote the ground truth label jth segment (namely jth input data) and its corresponding prediction result, respectively. Let N_{seg} be the total number of segments. Following the supervised algorithms in [7,17,19,42,43], this paper adopts the cross entropy loss function, $\mathcal{L}$, to compute the average difference between the ground truth labels and their corresponding prediction results, as written in Eq. (2).

$$\mathcal{L} = -\frac{1}{N_{seg}} \sum_{j=1}^{N_{seg}} y_j log(p_j) \tag{2}$$

3.2. HARFLS

HARFLS is a secure distributed system for human activity recognition. Let N_{tot} and N_{conn} denote the numbers of total and connected users in the system, respectively, where $N_{conn} \leq N_{tot}$. The architecture of HARFLS is shown in Fig. 3. Its core idea is to encourage users to train their models locally and share their model weights using the FedAvg method. To be specific, each user

trains its model by the data collected from the corresponding mobile device. The model weights are uploaded to a HARFLS server periodically. Once the weights from the connected users are received, the HARFLS server calculates the average weights and sends them back to all users. Each connected user initiates its next stage of training based on the average weights it receives from the server. To avoid model weights leakage, we apply encryption and decryption in the weights uploading and distribution processes, respectively. This system allows users to enjoy the benefits brought by model sharing without sacrificing privacy and security.

The following describes three practical concerns in HARFLS, including *security & privacy protection, communication overhead*, and *weight update*.

3.2.1. Security & privacy protection

HARFLS protects users' information in two ways. First, each connected user trains its model based on its own data and sends the model weights rather than the training data to the HARFLS server. This avoids exposing users' local data and reduces the bandwidth consumption in the network [34,35].

Second, to enhance the protection against weights leakage, homomorphic encryption, a well recognized encryption method [39], is applied before weights transmission over the network. This encryption method is provided by the PySyft[1] community and directly embedded into our system.

3.2.2. Communication overhead

Let *FLIs* be the number of federated learning iterations. In this paper, a federated learning iteration is defined as a complete weights uploading and dissemination cycle between users and the server. As aforementioned, the homomorphic encryption method is integrated into HARFLS. Compared with the bare transmission, encryption and decryption incur additional computational costs. For example, assume there are ten connected users in the HARFLS system. For a federated learning iteration, the average run time with and without encryption and decryption costs 7.8262 mins and 6.9808 mins, respectively (found in Table 1). The reason behind this is the encryption and decryption operations cause extra computational overhead.

HARFLS does not calculate and circulate the average weights until the model weights from the N_{conn} connected users are received [31,32,34]. This effectively reduces the interaction between these users and the server, which improves the service efficiency of the HARFLS system. For an arbitrary connected user, e.g., U_i, we analyze the communication cost of uploading its model weights. Let B be the bandwidth requirement for uploading the model weights of U_i once. Then, the total uplink communication cost of U_i is proportional to $B \cdot FLIs$. Any user requires the same bandwidth, i.e., B, during weights uploading since all users adopt an identical model structure although different models are associated with different weights. Hence, the total uplink communication cost of all the connected users is in proportion to $B \cdot FLIs \cdot N_{conn}$.

3.2.3. Weight update

As the previous works in [31–34] suggest, the FedAvg method is used for model converging. Denote the model of U_i by M_i. Let w_i^t and w_{sum}^t be the weights of M_i and the average weights in the tth federated learning iteration, respectively. w_{sum}^t is defined in Eq. (3).

$$w_{sum}^t = \frac{1}{N_{tot}} \sum_{i=1}^{N_{conn}} w_i^t, \qquad N_{conn} \leq N_{tot} \tag{3}$$

[1] https://github.com/OpenMined/PySyft.

Algorithm 1 HARFLS User Implementation Procedure

```
procedure UserProcedure(U_i, FLIs)
    Initialize all global variables;
    for t = 1 to FLIs do
        if t == 1 then
            Obtain w_i^t by the initial local training;
            Encrypt w_i^t and upload Encry(w_i^t);
        else
            if receiveServer(Active)==1 then
                // Connect to the HARFLS server
                Receive Encry(w_sum^t);
                Obtain w_sum^t by Decry(Encry(w_sum^t));
                // Replace w_i^t with w_sum^t
                Obtain w_i^{t+1} based on w_sum^t by Eq. (4);
                Encrypt w_i^{t+1} and upload Encry(w_i^{t+1});
            else
                Disconnect from the HARFLS server.
            end if
        end if
    end for
end procedure
```

Algorithm 2 HARFLS Server Implementation Procedure

```
procedure ServerProcedure(N_tot, N_conn, FLIs)
    Initialize all global variables;
    Set W = ∅;
    for t = 1 to FLIs do
        // Run on the server
        Clear and initialize W;
        for i = 1 to N_conn do
            // Receive model weights from users
            Receive Encry(w_i^t);
            Obtain w_i^t by Decry(Encry(w_i^t));
            Include w_i^t in W;
        end for
        Obtain w_sum^t based on W by Eq. (3);
        Encrypt w_sum^t and send to all users.
    end for
end procedure
```

Then, w_{sum}^t is sent to each user and used as the initial weights of M_i at the beginning of the next federated learning iteration. Let $\mathcal{L}$ denote the cross-entropy loss function (see Eq. (2) for details). The weights of M_i in the $(t+1)$-th federated learning iteration, w_i^{t+1}, is updated using Eq. (4).

$$w_i^{t+1} \leftarrow w_{sum}^t - \varepsilon_i \nabla_{w_{sum}^t} \mathcal{L}(w_{sum}^t) \tag{4}$$

where, ε_i is the learning rate of M_i, and $\nabla_{w_{sum}^t}$ is the gradient matrix of M_i when updating M_i by w_{sum}^t. Alg. 1 and Alg. 2 show the user and server implementation procedures, respectively.

4. Experiments

This section introduces the experimental setup and performance metrics and analyzes the experimental results.

4.1. Experimental setup

4.1.1. Dataset description

Following [22,24,42,43,48,50,62,63] , we select four widely used HAR datasets for performance evaluation, i.e., the wireless

Table 1
The run time (mins) taken for one federated learning iteration on a GTX-1080Ti with 11G on the WISDM dataset.

Algorithms	$N_{conn} = 10$	$N_{conn} = 20$	$N_{conn} = 30$	$N_{conn} = 40$	$N_{conn} = 50$	$N_{conn} = 100$
HARFLS	7.8262	9.7992	11.9499	14.0431	18.4699	28.9939
HARFLS without encryption and decryption	6.9808	8.8394	10.9342	12.0031	16.3683	26.2399

Table 2
Details of the four HAR datasets.

Datasets	Sample Rate	Activities	Classes	Samples
WISDM	20 Hz	Walking (Wk), Jogging (Jg), Upstairs (Us), Downstairs (Ds), Sitting (St), and Standing (Sd)	6	1,098,207
UCI_HAR2012	50 Hz	Walking (Wk), Upstairs (Us), Downstairs (Ds), Sitting (St), Standing (Sd), and Laying (Lay)	6	10,299
OPPORTUNITY	32 Hz	Standing (Sd), Walking (Wk), Sitting (St), and Lying (Ly)	4	701,366
PAMAP2	100 Hz	Lying (Ly), Sitting (St), Standing (Sd), Walking (Wk), Running (Run), Cycling (Cy), Nordic Walking (NW), Upstairs (Us), Downstair (Ds), Vacuum Cleaning (VC), and Ironing (Ir)	11	1,893,512

sensor data mining laboratory (WISDM), UCI HAR using smartphones in 2012 (UCI_HAR 2012), OPPORTUNITY, and PAMAP2 datasets, shown in Table 2. These datasets are used to verify the effectiveness of PEN by comparing it with 14 existing HAR algorithms. Meanwhile, we use two large datasets with millions of samples, i.e., WISDM and PAMAP2, to compare HARFLS with 11 FL systems in the literature.

4.1.2. Data preprocessing

For HAR problems, the collected raw samples are segmented into sequences based on a specific window size. These sequences carry necessary information about activities and are used as input data of recognition algorithms. Nowadays, segmentation methods adopt either fixed or adaptive window size [25,64]. Compared with the adaptive window size segmentation, the fixed window size segmentation is beneficial to end-to-end learning [22,23]. As the proposed PEN is an end-to-end supervised training model, this paper fixes the window size for segmentation.

Following the previous work in [22,23], we use the sliding window method to generate a series of segments as the input data to PEN, i.e., $\mathbf{S} = \{S_1, S_2, \ldots, S_{N_{seg}}\}$, where, N_{seg} is the total number of segments. Each segment, S_m, is a sequence of length K composed of the data collected by the wearable sensors, e.g., the triaxial accelerometers, defined in Eq (5).

$$S_m = [D_i, D_{i+1}, \ldots, D_{i+K-1}] \tag{5}$$

where, D_i represents the data collected in the ith sensing period.

4.1.3. Data partition

As suggested by [22–24], $\mathbf{S}$ is randomly divided into two parts, i.e., S_{train} for training and S_{test} for testing. S_{train} and S_{test} are used for model training and performance evaluation, respectively.

As the studies in [33,34,57,58] suggest, S_{train}, is randomly divided into N_{tot} parts, namely, $S_{train} = \{S_{train}^{(1)}, S_{train}^{(2)}, \ldots, S_{train}^{(N_{tot})}\}$, where $S_{train}^{(i)}$ is used as the local data of U_i for training M_i. We adopt the random partition rather than the average partition because the former is more appropriate for verifying the actual performance of a FL system [32,57]. The scikit-learn[2] software is used to randomly partition S_{train}. As for testing, S_{test} is completely assigned to each user, i.e., $S_{test}^{(i)} = S_{test}$, $i = 1, 2, \ldots, N_{tot}$. Let $N_{train}^{(i)}$ and $N_{test}^{(i)}$, $i = 1, 2, \ldots, N_{tot}$, represent the number of segments for training and testing M_i, respectively. Let N_{train} denote the number of segments in S_{train}. N_{seg} is defined in Eq. (6).

$$\begin{aligned} N_{seg} &= N_{train} + N_{test} \\ &= \sum_{i=1}^{N_{tot}} N_{train}^{(i)} + N_{test} \end{aligned} \tag{6}$$

where, N_{test} is the number of segments in S_{test}. Since all users utilize S_{test} as their testing dataset, the relationship between N_{test} and $N_{test}^{(i)}$, $i = 1, \ldots, N_{tot}$ is written in Eq. (7).

$$N_{test} = N_{test}^{(1)} = N_{test}^{(2)} = \cdots = N_{test}^{(N_{tot})} \tag{7}$$

4.1.4. Implementation details

We use the Kaiming initialization method [65] to initialize M_i and set all its bias values to zero, $i = 1, \ldots, N_{tot}$. Following the previous studies in [31,33], we apply stochastic gradient descent (SGD) to optimize M_i, where the momentum, weight decay and initial learning rate values are set to 0.9, 0.0001 and 0.001, respectively.

We run all experiments on a computer with Ubuntu 18.04 OS, an Nvidia GTX 1070Ti GPU with 8GB, an Nvidia GTX 1080Ti GPU with 11GB, and an AMD R5 1400 CPU with 16G RAM. All models are implemented using Python 3.6,[3] Pytorch 1.3[4] and PySyft.

4.2. Performance metrics

Following the previous studies in [22–24], we use the widely recognized *Accuracy* and F1-score (also called F-measure) to verify the classification performance of HARFLS, where F1-score is based on two metrics, namely *Precision* and *Recall*.

Accuracy is the percentage (%) the overall accuracy for all classes, defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \tag{8}$$

where, TP and TN are the numbers of true positive and negative segments, respectively. FP and FN denote the numbers of false positive and negative segments, respectively.

[2] https://scikit-learn.org/stable/.

[3] https://www.python.org/.

[4] https://pytorch.org/.

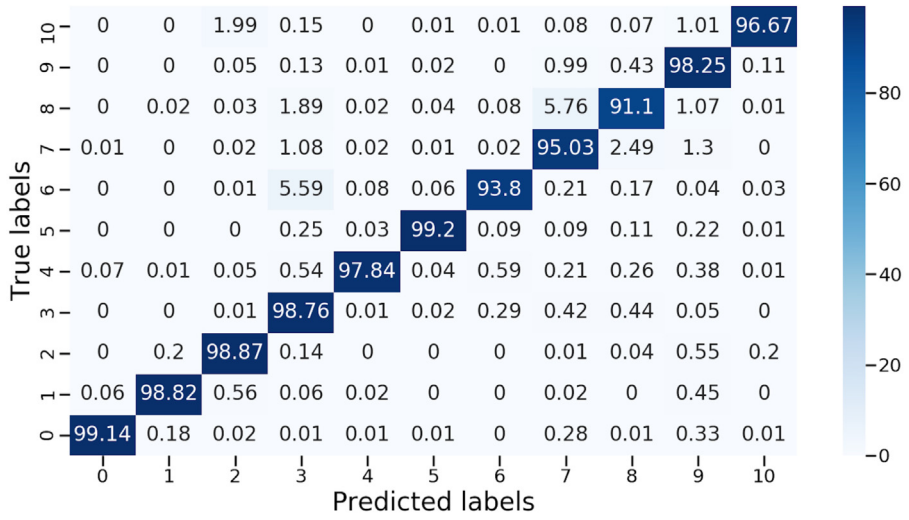


Fig. 4. Confusion matrix of PEN on the PAMAP2 testing set in terms of *Accuracy*.

Table 3
Results of F_1 obtained by the four structures on four HAR datasets (±1.83).

Datasets	FN	RN	PEN with *ReLU*	PEN
WISDM	92.26	82.19	96.68	**98.97**
UCI_HAR2012	93.14	83.43	95.23	**96.33**
PAMAP2	93.68	84.53	95.69	**97.78**
OPPORTUNITY	93.12	86.34	94.39	**96.89**
Mean F_1	93.05	84.12	95.50	**97.49**

Precision is the percentage (%) of the positive segments correctly classified to the total number of segments classified as positive, as expressed in Eq. (9).

$$Precision = \frac{TP}{TP + FP} \times 100\% \tag{9}$$

Recall is the percentage (%) of the positive segments correctly classified to the total number of actual positive segments, shown in Eq. (10).

$$Recall = \frac{TP}{TP + FN} \times 100\% \tag{10}$$

F1-score comprehensively trades off between *Precision* and *Recall*, as defined in Eq. (11).

$$F_1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{11}$$

4.3. Evaluation of PEN

This subsection first provides an ablation study on PEN and then compares PEN with various HAR algorithms and feature extractors for FL.

4.3.1. Ablation study

As shown in Fig. 1, PEN consists of a feature network and a relation network, with *LeakReLU* as the activation function. To evaluate the effectiveness of each component above, we compare the performance of four structures listed below on four HAR datasets.

- FN: the feature network of PEN.
- RN: the relation network of PEN.
- PEN with *ReLU*: PEN with *ReLU* instead of *LeakReLU*.
- PEN: the proposed feature extractor in this paper.

Table 3 shows the F_1 results obtained by FN, RN, PEN with *ReLU*, and PEN. If comparing PEN with FN and RN, one can easily see that PEN results in the highest mean F_1 value, i.e., 97.49%. This is because FN and RN focus on extracting local features and global relationships, respectively. But, none of them is capable of effectively capturing the two types of features above simultaneously as PEN does. Between PEN and PEN with *ReLU*, the former overweighs the latter because, unlike *ReLU*, *LeakReLU* is sensitive to negative numbers, reducing the loss of valuable information during the feature transmission process. This is why we adopt *LeakReLU* as the activation function for our PEN.

4.3.2. Comparison with existing HAR algorithms

To reflect the performance of the proposed PEN, we compare it with a number of existing HAR algorithms against F_1 on four HAR datasets. The results are collected in Table 4. Note that the methods in [62,63,66] are traditional methods while those in [22–24,50] are deep learning ones. It is easily seen that the deep learning methods obtain better F_1 results than the traditional ones. The method based on 1DCNN [22] is the worst deep learning method in terms of the mean F_1 value. Yet, it performs better than the method based on LR, the best-performance traditional method. The following explains why. The traditional methods are only able to identify significant differences in data and thus not capable of extracting complex features. In contrast, deep learning can explore the intrinsic correlations among representations, by constructing the internal representation hierarchy of data.

PEN, MCA, and CNNBiLSTM are the three best algorithms since they obtain a mean F_1 value of 97.49%, 95.50%, and 95.38%, respectively. The following explains the reasons. By taking advantage of its feature and relation networks, PEN mines plenty of meaningful representations from the HAR data. CNNBiLSTM cascades the CNN and BiLSTM networks for capturing potential information from HAR datasets. MCA utilizes the multi-head CNN and attention networks to explore multi-scale features from data, ensuring its HAR performance.

In addition, to verify the PEN's effectiveness, we compare it with a number of machine and deep learning algorithms in terms of the number of parameters and run time, on the PAMAP2 testing set. The results are collected in Table 5. The proposed PEN is slower than the four machine learning algorithms, i.e., SVM, KNN, BAGGING, and Random Forest. However, if compared with the six deep learning algorithms, PEN reflects certain advantages, e.g., its parameters on the PAMAP2 dataset is 373,260, which is lower than 1DCNN, 2DCNN, LSTM, CNNLSTM, and MCA. The

Table 4
Results of F_1 obtained by different HAR algorithms on four HAR datasets ($\pm$ 1.02). Abbreviations: STT – ST+Time, LR – Logistic Regression, MP – Multi-Perceptron, SDA – Stacked Denoising Autoencoder, MCA – Multi-head Convolutional Attention, and Lego-CNN – Lego Lightweight CNN.

Datasets	SVM [66]	BAGGING [63]	KNN [63]	STT [63]	J48 [62]	LR [62]	MP [62]	SDA [23]	1DCNN [22]	2DCNN [22]	CNNLSTM [50]	CNNBiLSTM [50]	MCA [22]	Lego-CNN [24]	PEN
WISDM	80.20	81.30	75.20	93.60	85.10	87.10	91.70	94.00	92.13	91.89	94.89	95.37	95.40	97.30	**98.97**
UCI_HAR2012	83.56	92.30	34.31	–	–	87.30	83.10	–	91.72	92.60	94.92	95.83	95.68	95.41	**96.33**
PAMAP2	86.49	83.28	85.34	–	–	86.59	75.78	–	92.32	92.35	93.16	95.12	95.55	93.06	**97.78**
OPPORTUNITY	87.59	88.63	86.24	–	–	87.49	89.68	–	92.89	93.24	94.62	95.19	95.37	92.21	**96.89**
Mean F_1	84.46	86.38	70.27	–	–	87.12	85.07	–	92.27	92.52	94.40	95.38	95.50	94.50	**97.49**

Table 5
The number of parameters and run time results with different algorithms on the PAMAP2 testing set.

Algorithms	Parameters	CPU Only (s)	With GPU 1070Ti (s)	With GPU 1080Ti (s)
SVM	–	6.8943	–	–
KNN	–	20.6835	–	–
BAGGING	–	13.5964	–	–
Random forest	–	16.3459	–	–
MP	89,792	7.9934	0.9345	0.8912
1DCNN	505,292	93.7594	5.0139	4.5128
2DCNN	1,157,132	261.5131	7.9969	7.4839
LSTM	808,716	18.7049	1.2113	1.2405
CNNLSTM	2,848,588	2009.8893	55.4653	54.9549
MCA	2,658,962	405.3406	11.3494	11.5694
PEN	373,260	45.5159	3.4515	3.6112

Table 6
Results of F_1 with different feature extractors on the WISDM and PAMAP2 datasets (±1.02).

Datasets	KNN [67]	SVM [67]	LR [67]	LSTM [56]	CNN [57]	CNN [58]	CNN [68]	MP [68]	1DCNN [22]	CNNLSTM [50]	MCA [22]	PEN
WISDM	78.87	81.34	84.75	82.24	87.89	86.27	88.63	87.18	92.13	94.89	95.40	**98.97**
PAMAP2	76.84	81.63	82.69	83.59	87.92	87.02	86.74	88.17	92.32	93.16	95.55	**97.78**
Mean F_1	77.86	81.49	83.72	82.92	87.91	86.65	87.69	87.68	92.23	94.03	95.48	**98.38**

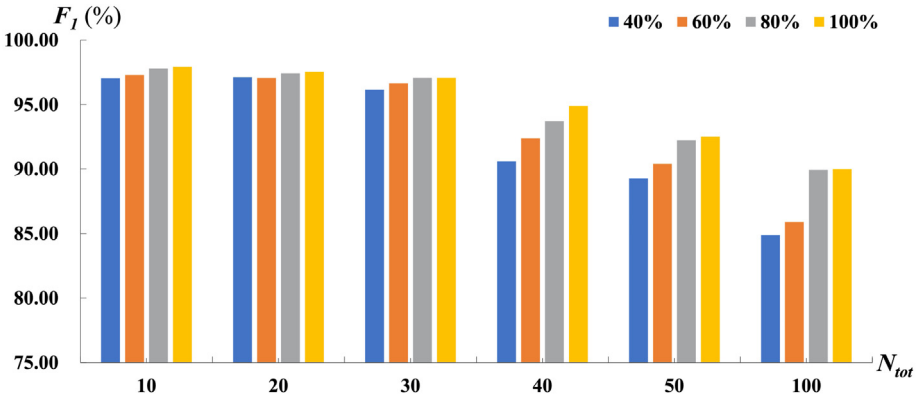


Fig. 5. Results obtained by HARFLS with different N_{tot} values and ratios of N_{conn} to N_{tot} on the WISDM dataset.

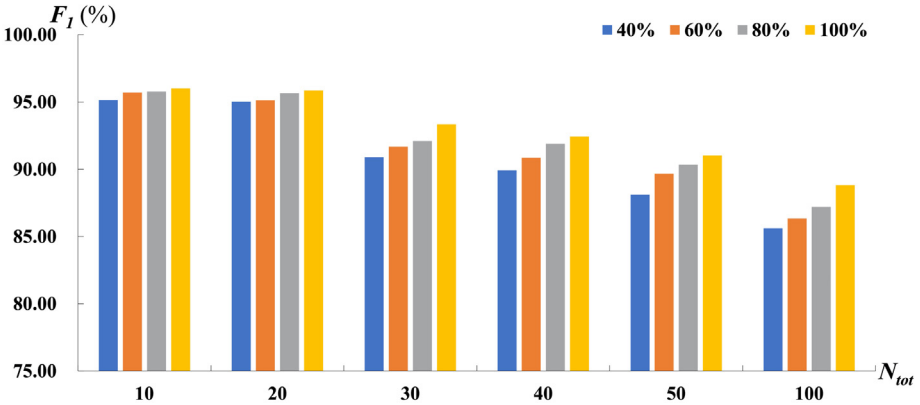


Fig. 6. Results obtained by HARFLS with different N_{tot} values and ratios of N_{conn} to N_{tot} on the PAMAP2 dataset.

results on the WISDM, UCI_HAR2012, and OPPORTUNITY datasets are collected in Appendix.

Besides, to visualize the performance of PEN, the authors show its confusion matrix on the PAMAP2 testing set in Fig. 4. Its confusion matrices on the WISDM, UCI_HAR2012, and OPPORTUNITY testing sets are in Appendix.

4.3.3. Comparison with existing feature extractors for FL

A feature extractor is crucial for an FL system since it is responsible for mining features from HAR data for each FL user. To further evaluate the performance of PEN, we compare it with a number of feature extractors in the existing FL systems on the WISDM and PAMAP2 datasets. These datasets contain millions of samples.

Table 7
Results of F_1 obtained by different FL systems with $N_{tot} = 100$ on the WISDM and PAMAP2 datasets ($\pm$ 1.41).

Authors	Methods	N_{conn}	WISDM (%)	PAMAP2 (%)
Mothukuri et al. [67]	FL with KNN	100	72.61	71.54
	FL with SVM	100	74.95	73.49
	FL with LR	100	75.43	76.34
Zhao et al. [56]	FL with LSTM	100	74.92	74.18
Tang et al. [57]	FL with CNN	100	80.02	81.36
Wu et al. [58]	FL with CNN	100	78.14	77.59
Dong et al. [68]	EaSTFLy with CNN	100	81.05	81.93
	EaSTFLy with MP	100	79.69	78.91
Ours	HARFLS with 1DCNN	100	83.23	82.13
	HARFLS with CNNLSTM	100	86.13	86.08
	HARFLS with MCA	100	87.93	87.02
	HARFLS	40	84.88	85.62
		60	85.89	86.34
		80	89.93	87.21
		100	**89.99**	**88.82**

Table A.1
The number of parameters and run time results with different algorithms on the WISDM testing set.

Algorithms	Parameters	CPU Only (s)	With GPU 1070Ti (s)	With GPU 1080Ti (s)
SVM	–	5.8365	–	–
KNN	–	9.4432	–	–
BAGGING	–	6.2953	–	–
Random forest	–	9.2659	–	–
MP	89,638	7.3983	1.9342	1.1658
1DCNN	772,960	19.0011	8.0058	8.2335
2DCNN	1,124,334	21.7324	8.4236	7.8746
LSTM	204,324	16.8543	5.1023	5.0002
CNNLSTM	2,837,317	69.6823	17.2394	17.2398
MCA	2,771,270	22.3498	11.9356	11.8345
PEN	223,558	17.9467	7.9308	7.4801

Table A.2
The number of parameters and run time results with different algorithms on the UCI_HAR2012 testing set.

Algorithms	Parameters	CPU Only (s)	With GPU 1070Ti (s)	With GPU 1080Ti (s)
SVM	–	0.2059	–	–
KNN	–	0.9346	–	–
BAGGING	–	1.1045	–	–
Random forest	–	3.4596	–	–
MP	164,396	1.0758	0.2034	0.1964
1DCNN	920,145	15.2322	2.8934	2.1262
2DCNN	1,155,719	31.6259	3.0031	2.8849
LSTM	1,364,487	3.1376	1.2342	0.9435
CNNLSTM	2,913,863	37.7875	7.1346	7.0921
MCA	2,896,913	35.8435	6.3452	5.8934
PEN	819,911	7.4029	1.5347	1.4934

Table A.3
The number of parameters and run time results with different algorithms on the OPPORTUNITY testing set.

Algorithms	Parameters	CPU Only (s)	With GPU 1070Ti (s)	With GPU 1080Ti (s)
SVM	–	3.0134	–	–
KNN	–	60.4932	–	–
BAGGING	–	15.4325	–	–
Random Forest	–	70.4532	–	–
MP	113,588	3.6123	0.5005	0.4837
1DCNN	503,237	499.3542	14.5934	14.4923
2DCNN	1,155,333	1424.0013	29.7471	28.9934
LSTM	1,037,317	4.8367	0.5466	0.5331
CNNLSTM	2,723,890	1299.0964	36.7665	37.2133
MCA	2,493,860	908.4923	22.3423	21.9934
PEN	998,081	190.7873	7.4985	7.4331

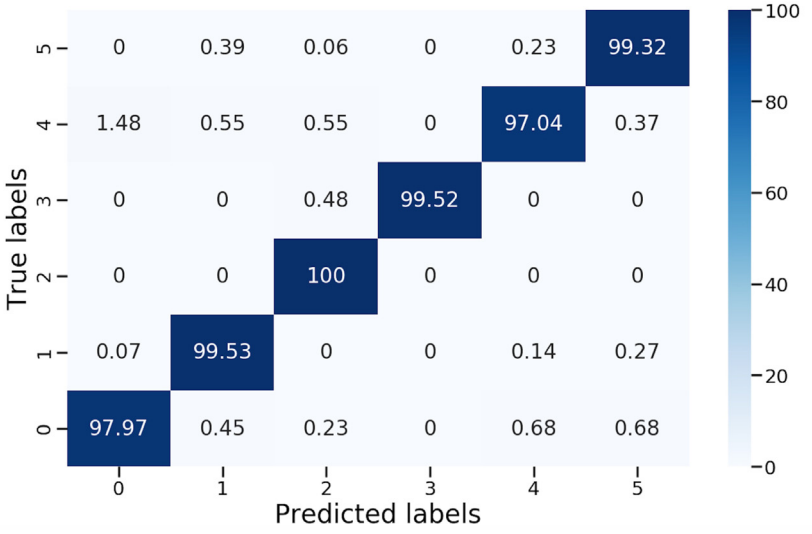


Fig. 7. Confusion matrix of PEN on the WISDM testing set in terms of the *Accuracy*.



Fig. 8. Confusion matrix of PEN on the UCI_HAR2012 testing set in terms of the *Accuracy*.

Table 6 shows the F_1 results obtained by different feature extractors on the WISDM and PAMAP2 datasets. Apparently, PEN achieves the best performance as its mean F_1 value is the highest, namely, 98.38%. This is because the features extracted by the feature network and those by the relation network complement each other. MCA obtains decent results as the multi-head CNN and attention networks help to mine rich features from the HAR data. In contrast, the traditional feature extractors are more likely to lead to unsatisfactory performance, e.g., KNN only obtains a mean F_1 value of 77.86%. The reason is that these extractors are good at exploring shallow features rather than hierarchical representations from sequential data.

4.4. Evaluation of HARFLS

This subsection first investigates the impacts of N_{tot} and N_{conn} on the HARFLS performance and then compares HARFLS with a number of existing FL systems in the literature.

4.4.1. Impacts of N_{tot} and N_{conn}

To study the trade-off between HAR performance and system capacity, we test our HARFLS with six different N_{tot} values, including 10, 20, 30, 40, 50, and 100 on the WISDM and PAMAP2 datasets. We consider four different ratios of N_{conn} to N_{tot} for each N_{tot}, namely 40%, 60%, 80%, and 100%. For example, if there are 10 users in the system, i.e., $N_{tot} = 10$, 40% means there are 4 connected users for weights uploading.

The results of F_1 obtained by HARFLS with different N_{tot} and N_{conn} values on the WISDM and PAMAP2 datasets are shown in Figs. 5 and 6, respectively. First of all, we study the impact of N_{conn} on the HAR performance. For each N_{tot}, a larger N_{conn} tends to result in a higher F_1 value. For instance, when $N_{tot} = 10$, 4, 6, 8 and 10 connected users lead to 97.03%, 97.29%, 97.79%, and 97.92% on the WISDM dataset, respectively. This is because as N_{conn} increases, more motion data are used by the system and thus more useful information can be mined from the data.

Secondly, we focus on the impact of N_{tot} on the performance of HARFLS. For each ratio of N_{conn} to N_{tot}, a larger N_{tot} is likely to result in a lower F_1 value. For instance, when the ratio of N_{conn} to N_{tot} is set to 60% on the WISDM dataset, the F_1 values obtained

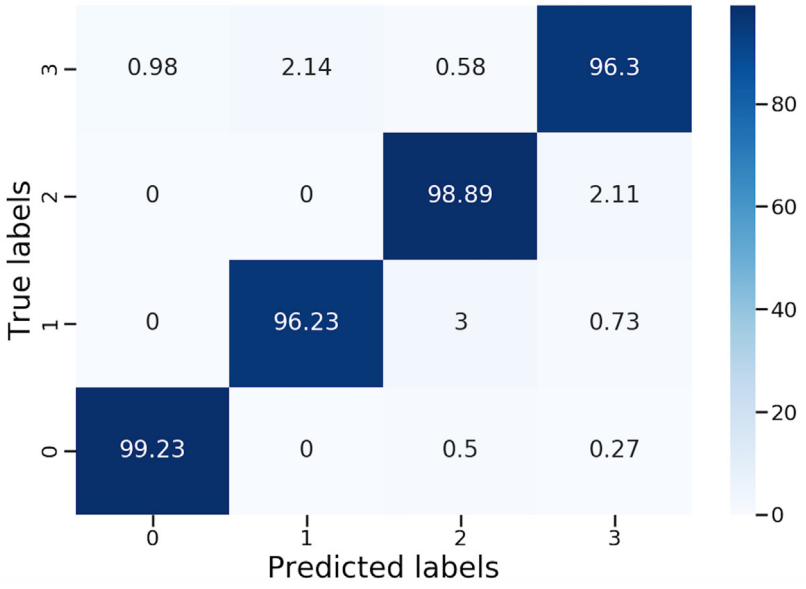


Fig. 9. Confusion matrix of PEN on the OPPORTUNITY testing set in terms of the *Accuracy*.

with N_{tot} = 10, 20, 30, 40, 50, and 100, are 97.29%, 97.06%, 96.65%, 92.38%, 90.39%, and 85.89%, respectively. If we fix the ratio of N_{conn} to N_{tot}, N_{tot} is proportional to N_{conn}. More users in the system means more users connect to the server and participate in the training process, bringing more uncertainties to the system. This, to some extent, deteriorates the system's performance.

When focusing on results on WISDM, one can see that as N_{tot} increases from 10 to 30, the F_1 value is slightly weakened; as N_{tot} rises from 30 to 100, the F_1 value drops quickly. It looks like HARFLS with $N_{tot} = 30$ achieves a decent trade-off between HAR accuracy and system capacity. When N_{tot} reaches 40 and beyond, significant performance degradation results. It is obvious that the system performance trend on PAMAP2 is similar to that on WISDM as the number of connected users increases. Specifically, with more users connected to the system, the system's performance becomes worse. As N_{tot} increases from 10 to 20, the F_1 value slightly decreases; as N_{tot} grows up from 20 to 100, the F_1 value goes down quickly. HARFLS with $N_{tot} = 20$ achieves a good compromise between HAR accuracy and system capacity. When N_{tot} is larger than 20, the system suffers from significant performance loss.

4.4.2. Comparison with existing FL systems

To further evaluate the performance of HARFLS, we compare it with 11 existing FL systems. These systems include three FL systems based on traditional algorithms (i.e., FL with KNN, FL with SVM, and FL with LR [67]), three FL systems previously developed for tackling HAR problems (i.e., FL with LSTM [56], FL with CNN [57], and FL with CNN [58]), two EASTFLY systems (EaSTFLy with CNN and EaSTFLy with MP [68]), and three HARFLS systems(i.e., HARFLS with 1DCNN, HARFLS with MCA, and HARFLS with CNNLSTM). In the literature, an FL system with $N_{tot} = 100$ is considered as a relatively large scenario [56–58,67,68]. So, we set $N_{tot} = 100$ in this experiment. For the existing FL systems, we fix N_{conn} to 100. By allowing all users to participate in the system training process, each existing FL system should yield its best performance. In the meantime, we test our system with different ratios of N_{conn} to N_{tot}, namely 40%, 60%, 80%, and 100%. Table 7 shows the results of F_1 obtained by different FL systems on the WISDM and PAMAP2 datasets.

It is evident that our HARFLS achieves the highest F_1 values on WISDM and PAMAP2. Even if N_{conn} = 80, HARFLS still obtains 89.93% and 87.21%, better than the other FL systems, respectively. This is because the proposed PEN is capable of capturing sufficient temporal features from the HAR data for each user, hence the local models are relatively accurate. HARFLS with MCA ranks second on both the WISDM and PAMAP2 datasets, thanks to the multi-head CNN and attention networks responsible for feature extraction. On the other hand, the systems with traditional methods [67], including KNN, SVM, and LR, result in worse F_1 values than the other systems, e.g., FL with KNN only leads to 72.61% on the WISDM dataset. The reason is that traditional methods are sensitive to the significant differences rather than the intrinsic and multi-scaled structures in the HAR data.

5. Conclusion

This paper presents HARFLS, a secure federated learning system for human activity recognition (HAR). This system adopts PEN as its feature extractor and the FedAvg method for model weights sharing, respectively. The feature and relation networks effectively explore local features and global relationships for PEN. Experimental results show that PEN obtains the highest $F1$ results among all HAR algorithms and FL feature extractors on four widely recognized datasets. Furthermore, HARFLS obtains a decent trade-off between recognition accuracy and system capacity when $N_{tot} = 30$ and $N_{tot} = 20$ on the WISDM and PAMAP2 datasets, respectively. Thanks to PEN's strong feature extraction ability, our HARFLS performs better than 11 other FL systems regarding the recognition accuracy, even if the ratio of N_{conn} to N_{tot} is 80% on the WISDM and PAMAP2 datasets. This reflects the potential of HARFLS to be applied to real-world HAR applications.

CRediT authorship contribution statement

Zhiwen Xiao: Methodology, Conceptualization, writing – original draft. **Xin Xu:** Methodology, Software. **Huanlai Xing:** Methodology, writing – review & editing, Supervision. **Fuhong Song:** Software, Validation. **Xinhan Wang:** Investigation, Visualization. **Bowen Zhao:** Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

See Figs. 7–9 and Tables A.1–A.3.

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